

size

Array size

Syntax

```
sz = size(A)
szdim = size(A,dim)
szdim = size(A,dim1,dim2,...,dimN)
[sz1,...,szN] = size( __ )
```

Description

`sz = size(A)` returns a row vector whose elements are the lengths of the corresponding dimensions of A. For example, if A is a 3-by-4 matrix, then `size(A)` returns the vector [3 4]. [example](#)

If A is a table or timetable, then `size(A)` returns a two-element row vector consisting of the number of rows and the number of table variables.

`szdim = size(A,dim)` returns the length of dimension `dim` when `dim` is a positive integer scalar. You can also specify `dim` as a vector of positive integers to query multiple dimension lengths at a time. For example, `size(A,[2 3])` returns the lengths of the second and third dimensions of A in the 1-by-2 row vector `szdim`. [example](#)

`szdim = size(A,dim1,dim2,...,dimN)` returns the lengths of dimensions `dim1,dim2,...,dimN` in the row vector `szdim`. [example](#)

`[sz1,...,szN] = size( __ )` returns the lengths of the queried dimensions of A separately. [example](#)

Examples

[collapse all](#)

Size of 4-D Array

Create a random 4-D array and return its size.

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```
A = rand(2,3,4,5);
sz = size(A)
```

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```
sz = 1×4
```

```
     2     3     4     5
```

Query the length of the second dimension of A.

```
szdim2 = size(A,2)
```

[Get](#) ▼

```
szdim2 =
3
```

Query the length of the last dimension of A.

```
szdimlast = size(A,ndims(A))
```

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```
szdimlast =
5
```

You can query multiple dimension lengths at a time by specifying a vector dimension argument. For example, find the lengths of the first and third dimensions of A.

```
szdim13 = size(A,[1 3])
```

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```
szdim13 = 1×2
```

```
2    4
```

Find the lengths of the second through fourth dimensions of A.

```
szdim23 = size(A,2:4)
```

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```
szdim23 = 1×3
```

```
3    4    5
```

Alternatively, you can list the queried dimensions as separate input arguments.

```
szdim23 = size(A,2,3,4);
```

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Size of Table

Create a table with 5 rows and 4 variables.

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```
LastName = {'Smith';'Johnson';'Williams';'Jones';'Brown'};
Age = [38;43;38;40;49];
Height = [71;69;64;67;64];
Weight = [176;163;131;133;119];
BloodPressure = [124 93; 109 77; 125 83; 117 75; 122 80];
```

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```
A = table(Age,Height,Weight,BloodPressure,'RowNames',LastName)
```

A=5×4 table

	Age	Height	Weight	BloodPressure	
	—	—	—	—	
Smith	38	71	176	124	93
Johnson	43	69	163	109	77

Williams	38	64	131	125	83
Jones	40	67	133	117	75
Brown	49	64	119	122	80

Find the size of the table. Although the BloodPressure variable contains two columns, size only counts the number of variables.

```
sz = size(A)
```

 Get ▾

```
sz = 1x2
```

```
5    4
```

Dimension Lengths as Separate Arguments

Create a random matrix and return the number of rows and columns separately.

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```
A = rand(4,3);  
[numRows,numCols] = size(A)
```

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```
numRows =  
4  
numCols =  
3
```

Input Arguments

[collapse all](#)

A — Input array

scalar | vector | matrix | multidimensional array

Input array, specified as a scalar, a vector, a matrix, or a multidimensional array.

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64 | logical | char | string | struct | function_handle | cell | categorical | datetime | duration | calendarDuration | table | timetable

Complex Number Support: Yes

dim — Queried dimensions

positive integer scalar | vector of positive integer scalars | empty array

Queried dimensions, specified as a positive integer scalar, a vector of positive integer scalars, or an empty array of size 0-by-0, 0-by-1, or 1-by-0. If an element of dim is larger than ndims(A), then size returns 1 in the corresponding element of the output. If dim is an empty array, then size returns a 1-by-0 empty array.

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

dim1,dim2,...,dimN — List of queried dimensions

positive integer scalars

List of queried dimensions, specified as positive integer scalars separated by commas. If an element of the list is larger than `ndims(A)`, then `size` returns 1 in the corresponding element of the output.

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

Output Arguments

[collapse all](#)

sz — Array size

row vector of nonnegative integers

Array size, returned as a row vector of nonnegative integers.

- Each element of `sz` represents the length of the corresponding dimension of `A`. If any element of `sz` is equal to 0, then `A` is an empty array.
- If `A` is a scalar, then `sz` is the row vector `[1 1]`.
- If `A` is a table or timetable, then `sz` is a two-element row vector containing the number of rows and the number of variables. Multiple columns within a single variable are not counted.
- If `A` is a character vector of type `char`, then `size` returns the row vector `[1 M]` where `M` is the number of characters. However, if `A` is a string scalar, `size` returns `[1 1]` because it is a single element of a string array. For example, compare the output of `size` for a character vector and string:

```
szchar = size('mytext')
```

```
szchar =
```

```
1     6
```

```
szstr = size("mytext")
```

```
szstr =
```

```
1     1
```

To find the number of characters in a string, use the [strlength](#) function.

Data Types: double

szdim — Dimension lengths

nonnegative integer scalar | vector of nonnegative integer scalars | 1-by-0 empty array

Dimension lengths, returned as a nonnegative integer scalar when `dim` is a positive integer scalar, a row vector of nonnegative integer scalars when `dim` is a vector of positive integers, or a 1-by-0 empty array when `dim` is an empty array. If an element of the specified dimension argument is larger than `ndims(A)`, then `size` returns 1 in the corresponding element of `szdim`.

Data Types: double**sz1, ..., szN — Dimension lengths listed separately**
nonnegative integer scalars

Dimension lengths listed separately, returned as nonnegative integer scalars separated by commas.

- When `dim` is not specified and fewer than `ndims(A)` output arguments are listed, then all remaining dimension lengths are collapsed into the last argument in the list. For example, if `A` is a 3-D array with size `[3 4 5]`, then `[sz1,sz2] = size(A)` returns `sz1 = 3` and `sz2 = 20`.
- When `dim` is specified, the number of output arguments must equal the number of queried dimensions.
- If you specify more than `ndims(A)` output arguments, then the extra trailing arguments are returned as 1.

Data Types: double

Tips

- To determine if an array is empty, a scalar, or a matrix, use the functions [isempty](#), [isscalar](#), and [ismatrix](#). You can also determine the orientation of a vector with the [isrow](#) and [iscolumn](#) functions.

Extended Capabilities

[expand all](#)

Tall Arrays

Calculate with arrays that have more rows than fit in memory.

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Version History

Introduced before R2006a

[expand all](#)**R2019b:** Specify dimensions as vector of positive integers or separate input arguments

See Also

[length](#) | [strlength](#) | [ndims](#) | [numel](#) | [height](#) | [width](#)

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